Important Points

Engineering Chemistry (UE26CY151A) · Semester 1 · Unit 1 · Compiled from PESU AV-summary transcripts, class slides, the Unit-1 slide PDF and past ESA papers · 13 Oct 2026

★★★ high · ★★ medium · ★ low/conceptual weight | Not on current slides Transcript only Added for clarity | source tags: [S004·8] slide, [T005] transcript, [PDF] 137-page deck, [ESA] past paper

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M1 · Molecular Spectroscopy

T1 · EM radiation & interaction with matter

Definition: Molecular spectroscopy: the branch of science dealing with the interaction of electromagnetic radiation with matter (molecules). [S001·3] ★★

Spectrophotometer: source → monochromator (entrance slit, dispersive element, exit slit) → sample → detector → recorder. [S001·3] ★★

EM radiation has dual character (photons moving as waves); E and B fields oscillate perpendicular to each other and to the direction of propagation; all travel at the speed of light. [S001·4] ★★

Atomic spectra: electronic transitions between atomic orbitals → line spectra. Molecular spectra: electronic (HOMO→LUMO) plus vibrational and rotational transitions → band spectra. [S001·6-7] ★★

Absorption: molecule goes to a higher level taking ΔE = hν; emission: falls to a lower level releasing hν. [S001] ★

T2 · Electromagnetic spectrum & regions

Radio (10 m–1 cm): NMR, ESR – nuclear / electron spin reversal. Microwave (1 cm–100 μm): rotational. IR (100 μm–1 μm): vibrational. UV-Vis (1 μm–10 nm): electronic, valence electrons. X-ray (10 nm–100 pm): inner electrons. γ (100–1 pm): nuclear rearrangement. [S002·4-5] ★★★

Why: Rayleigh scattering: scattered light has the same frequency as the incident light (elastic). Raman: shifted frequency — Stokes lines (lower frequency, molecule gains energy), anti-Stokes lines (higher frequency, molecule loses energy). [T002] Transcript only Added for clarity ★★

Past-paper questions

(×2) Rayleigh vs Raman; Stokes vs anti-Stokes.

T3 · Born–Oppenheimer approximation

Statement: *Born–Oppenheimer approximation: electronic, vibrational and rotational energies of a molecule are independent; E = E_rot + E_vib + E_elec (E_trans negligible).* [S003·4] ★★

E_rot ≪ E_vib ≪ E_elec because electronic levels are farther apart than vibrational, which are farther than rotational. [S003·4] ★★

Exam trap: Mnemonic: Rot < Vib < Elec ↔ Microwave < IR < UV-Vis. [study aid] Added for clarity ★

Past-paper questions

(×3) State Born–Oppenheimer; energy-level diagram.

T3b · Beer–Lambert law

> Beer–Lambert law: absorbance $A = \log(I_0/I) = \varepsilon c l$ — proportional to concentration c and path length l; ε is the molar absorptivity. [ESA Dec-24] Not on current slides Added for clarity ★

Past-paper questions

Dec-24 (older syllabus) State the Beer–Lambert law.

T4 · Rotational (microwave) spectroscopy

> **Definition:** Gross selection rule (rotational): molecule must have a permanent dipole moment. Specific rule: $\Delta J = \pm 1$. [S004·3,10] ★★★

Why: HCl and CO are microwave-active (permanent dipole); H_2, Cl_2 and CO_2 are not (no permanent dipole). ★★★ [S004·5]

> Rigid-rotor model: heteronuclear diatomic = two masses joined by a rigid massless rod; J = 0,1,2…; levels 0, 2B, 6B, 12B… [S004·6-9] ★★★

> First line is at 2B (not B) and consecutive lines are 2B apart; B → I → r_0 via $I = \mu r_0^2$. [S004·12] ★★★

> Intensity rises then falls with J because higher levels are populated at room temperature (thermal population). [S004·12] ★★

Exam trap: Traps: J starts at 0; first line = 2B; use μ in kg (not amu); use c in cm s^{-1} when B is in cm^{-1}. ★★★ [exam]

Past-paper questions

2023–25 (×4) Derive $I = \mu r_0^2$ for a diatomic rigid rotor; calculate bond length from first line (HCl 21.18 cm^{-1}; HI 13.022 cm^{-1}); draw levels to J=5/6 and the spectrum.

T5 · Vibrational (IR) – SHO

> **Definition:** Gross selection rule (IR): the vibration must change the dipole moment; $\Delta v = \pm 1$. [S005·3,9] ★★★

Why: CO_2 has no permanent dipole (microwave-inactive) but is IR-active: asymmetric stretch and bending change the dipole; symmetric stretch does not. ★★★ [S005·4-5]

Why: HI is IR-active but H_2 and I_2 are not (homonuclear: no dipole change). ★★ [S005·4]

Why: SHO shows only one line: levels are equally spaced and $\Delta v = \pm 1$, so every allowed transition has the same energy $\bar{\nu}_{osc}$. ★★★ [S005·9]

> Zero-point energy $\varepsilon_0 = \frac{1}{2}\bar{\nu}_{osc} \neq 0$: the molecule never rests even at v = 0. Bond treated as a spring obeying Hooke's law → parabolic PE. [S005·6-9] ★★★

Every bond has a characteristic frequency, so an IR spectrum is a molecular 'fingerprint'. [S005·6] ★

Past-paper questions

2023–25 Calculate $\bar{\nu}$ and ZPE from k (HF, NO); k from $\bar{\nu}$ (CO); why only one line in SHO?

T6 · Vibrational – anharmonic

Why: Why SHO fails: real bonds dissociate when stretched; the true potential is anharmonic (Morse), so levels converge and overtones appear. [S006·3] ★★

Anharmonic: $\Delta v = \pm1, \pm2, \pm3\ldots$; most transitions start from v = 0 → fundamental (0→1), first overtone (0→2), second overtone (0→3). [S006·6] ★★

Hot bands: from v = 1 at higher temperature; 1→2 appears to the left of the fundamental. [S006·9] ★★

x_e is small and positive; line spacing decreases with increasing v. [S006·4] ★★

Past-paper questions

2023–25 Find x_e and $\omega\bar{}_e$ from fundamental and overtone; selection rule; failure of SHO.

T7 · Electronic spectroscopy & Franck–Condon

Electronic transitions: HOMO→LUMO ($\sigma\to\sigma^*$, $\pi\to\pi^*$, $n\to\pi^*$) in the UV-Vis; all molecules show them; visible transitions give colour. [S007·3] ★★★

Vibrational changes give 'coarse structure', rotational changes give 'fine structure'. Homonuclear molecules (H_2, N_2) show vibration/rotation only through electronic spectra. [S007·5] ★★

No vibrational selection rule during electronic transitions; most start at $v'' = 0$; lines labelled (v′,v″) form a progression converging at high energy (anharmonic excited state). [S007·6-8] ★★★

Statement: *Franck–Condon principle: "An electronic transition takes place so rapidly that a vibrating molecule does not change its internuclear distance appreciably during the transition." Hence transitions are vertical.* [S007·13] ★★★

Case (a) $r_e' = r_e''$: (0,0) strongest. Case (b) $r_e' > r_e''$: strongest line (v′,0) with v′>0. Case (c) $r_e' \gg r_e''$: transition goes beyond the dissociation limit → continuum. [S007·15-16] ★★★

Why: A continuum means dissociation: the atoms fly apart with unquantised kinetic energy. The excited state usually has a longer bond length because of antibonding character. [S007·10,16] ★★★

Past-paper questions

2023–25 (×5) State the Franck–Condon principle; draw the three cases with spectra; meaning of a continuum.

M2 · Phase Equilibria

T8 · Basic terms & phase rule

Definition: Phase (P): homogeneous, physically distinct, mechanically separable part of a system bounded by a surface. Gases: P = 1; miscible liquids: P = 1; each solid / polymorph is a phase. [S008·4] ★★★

Definition: Component (C): smallest number of independently variable constituents by which the composition of every phase can be expressed (zero/negative values allowed). [S008·6] ★★★

Definition: Degrees of freedom (F): minimum number of intensive variables (T, P, concentration) to be fixed to define the system completely. [S008·10] ★★★

Statement: *Gibbs phase rule: provided equilibrium is not influenced by external forces (gravity, electrical, magnetic), $F = C - P + 2$.* [S008·12] ★★★

Counting: ice/water/vapour P=3, C=1; $CaCO_3 \rightleftharpoons CaO + CO_2$: P=3, C=2; $NH_4Cl \rightleftharpoons NH_3 + HCl$: P=2, C=1 (C=2 if extra HCl/NH_3 added); $Na_2SO_4 \cdot 10H_2O$ system C=2. [S008·5-9] ★★

Systems minimise Gibbs energy at constant T, P; at equilibrium G is equal in coexisting phases. [S008·3] ★

Past-paper questions

(×2) Components and phases for $CaCO_3/MgCO_3$ decomposition.

T9 · Derivation of phase rule

At equilibrium all phases have the same T and P and the chemical potential of each component is equal in every phase; mass flows from higher to lower μ. [S009·3-4] ★★★

Derivation: variables = P(C−1) composition terms + 2 (T,P); constraints = C(P−1); F = C−P+2. [S009·7] ★★★

Past-paper questions

2023–25 (×3) Derive the phase rule for C components and P phases.

T10 · One-component system: water

Water: F = 3−P → one phase F=2 (bivariant area), two phases F=1 (univariant line), three phases F=0 (invariant point). [S009·8-10] ★★★

Triple point O: 0.0098 °C, 4.58 mmHg (F=0). Critical point A: 374 °C, 218 atm; above it water cannot be liquid. [S009·15] ★★★

Why: Fusion curve OC has negative slope: ice melts with decrease in volume ($\Delta V<0 \rightarrow dP/dT<0$), and it is steep because $|\Delta V|$ is small. Sublimation (OB) and vaporisation (OA) slopes are positive and less steep because ΔV is large. [S009·12-14; T009b] Transcript-assisted ★★★

Metastable OA′: supercooled water; its vapour pressure is higher than that of ice at the same T; disturbance returns the system to stable OB. [S009·16] ★★★

Past-paper questions

2023–25 (×4) Draw and label the water phase diagram; F in each region; metastable curve; negative slope of melting curve.

T11 · Two-component: reduced phase rule & Pb–Ag

Why: General phase rule not used for 2 components: C=2 gives F=4−P, F=3 for P=1, needing 3-D; so pressure is fixed (≈1 atm) → F = C−P+1. [S010·2-3] ★★★

Definition: Condensed system: gas phase absent, hardly affected by small pressure changes. [S010·3] ★★

Pb–Ag: components miscible in all proportions in melt, immiscible as solids; plot is temperature vs composition at constant pressure. [S010·4] ★★★

Freezing-point curves: C=2, P=2 → F=1 (univariant). Eutectic point O: P=3, F=0 (invariant), lowest freezing point, definite composition and sharp melting point. [S010·7-8] ★★★

Numerical facts: Pb mp 327 °C, Ag mp 961 °C, eutectic 303 °C at 97.4 % Pb / 2.6 % Ag. [S010; PDF] ★★★

Desilverisation of argentiferous lead (Pattinson-type process) exploits this: Pb crystallises first, leaving Ag-rich melt. [T010] Transcript only ★

Past-paper questions

2023–25 Draw the Pb–Ag diagram with eutectic values; why F = C−P+1.

T12 · Thermal analysis / cooling curves

Definition: Thermal analysis: study of cooling curves (temperature vs time) of various compositions during solidification; freezing and eutectic points are read from them. [S012·2] ★★

Pure substance: smooth fall, then a flat halt at the freezing point. Mixture: a change of slope when crystallisation starts, then a halt at the eutectic temperature. Plotting break/halt temperatures against composition builds the phase diagram. [S012·2-3] ★★

M3 · Computational Chemistry

T13 · Computational chemistry

Definition: Computational chemistry: techniques to solve chemical problems by simulating chemical systems, giving atomic-level information, using theoretical chemistry in efficient programs. [S011·3] ★

First theoretical work: Heitler and London, 1927. (Nobel 1998 Kohn & Pople; 2013 Karplus, Levitt & Warshel.) [S011·3; brief] Added for clarity ★

Scope: molecular properties (HOMO/LUMO, UV/IR/NMR spectra), reaction mechanism and kinetics, material design, drug discovery. [S011·6] ★

Classical: molecular mechanics (balls and springs, very fast) and molecular dynamics (motion of atoms). Quantum: semi-empirical, ab initio (wavefunction) and DFT (electron density; faster than ab initio, slower than semi-empirical). [S011·7-12] ★

Advantages: easier, cheaper, safer than experiments; complements, not replaces, experiment. [S011·13] ★

Ethylene: sp^2 C; σ + π bond; HOMO = π, LUMO = π*. Steps: geometry → basis set → method → solve → visualise. [S011·14-17] ★

AI-enhanced: fast screening, property prediction, reaction-pathway mapping, structure generation; speed-up from hours/days to milliseconds after training. [S011·19-20] ★

Past-paper questions

(×1) Define computational chemistry; three applications.